

Venue Activity Prediction: a model for Social Media Integration (working title)

Your name

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Abstract

Placeholder. Write the abstract last, once the model and the simulation results exist.

1 Introduction

2 Literature Review

3 Problem Statement

A social app lets users post a check-in at a public venue. A check-in records one drink, so one visit can produce several check-ins. Each check-in carries a user, a timestamp, and a venue. The app accepts a check-in only at the venue's own location. It serves a community of young adults who want to socialize with friends. We treat the community as a finite set, fixed only at the moment the app judges it, and its size can differ from one night to the next. We call a member who uses the app an *adopter*, and we call the percentage of the community who are adopters the *adoption rate*. The app also holds a social graph of mutual follows among adopters.

Assumption 1 (No non-adopter signal). *Non-adopters post nothing.*

Assumption 2 (Adopter- and venue-dependent posting probability). *A present adopter posts at least once with positive probability. That probability can depend on the adopter and on the venue.*

Assumption 3 (Friends at the same venue and time). *Non-friends post independently of one another. Friends at the same venue and time do not.*

Definition 1 (True headcount and occupancy). *Each venue holds a definite number of community members at any moment. We call that number its true headcount. Each venue also has a fixed capacity, and we call headcount divided by capacity its true occupancy. The app never reports either number.*

The app judges each venue from check-ins posted there in the last two hours, showing every venue with at least one such check-in. Each shown venue gets one of three levels: quiet (yellow), busy (orange), or packed (red). The two boundaries are the terciles of tonight's occupancy distribution among open venues, not a fixed historical threshold. Users compare levels, not numbers.

Definition 2 (Match rate and coverage). *The app gets a shown venue right when its level matches its true occupancy. The fraction of shown venues it gets right is the match rate. A venue that never appears is a different kind of error. The fraction of busy-or-packed venues that appear at all is the coverage.*

Our question is how match rate and coverage depend on adoption rate. We hold four other quantities fixed: the number of venues, the number of members out, how unequally they spread across venues, and how often an adopter posts. We ask for the smallest adoption rate at which match rate and coverage both reach 80 percent, and whether that adoption rate exists at all.

We judge match rate against every possible procedure, since the app could compute its levels in more than one way. Coverage needs no such care: a venue's appearance depends only on whether anyone posted, not on the procedure.

Real check-in data exists, but we do not use it. This paper's aim is to find the minimal adoption rate any algorithm needs before the app implements one, and no such algorithm exists yet to test. We keep real venue data out of scope too, and we set aside what the map does to the people who read it: every statement here describes a world where nobody has seen the map.

4 Model

5 Simulation

6 Results

7 Discussion

References